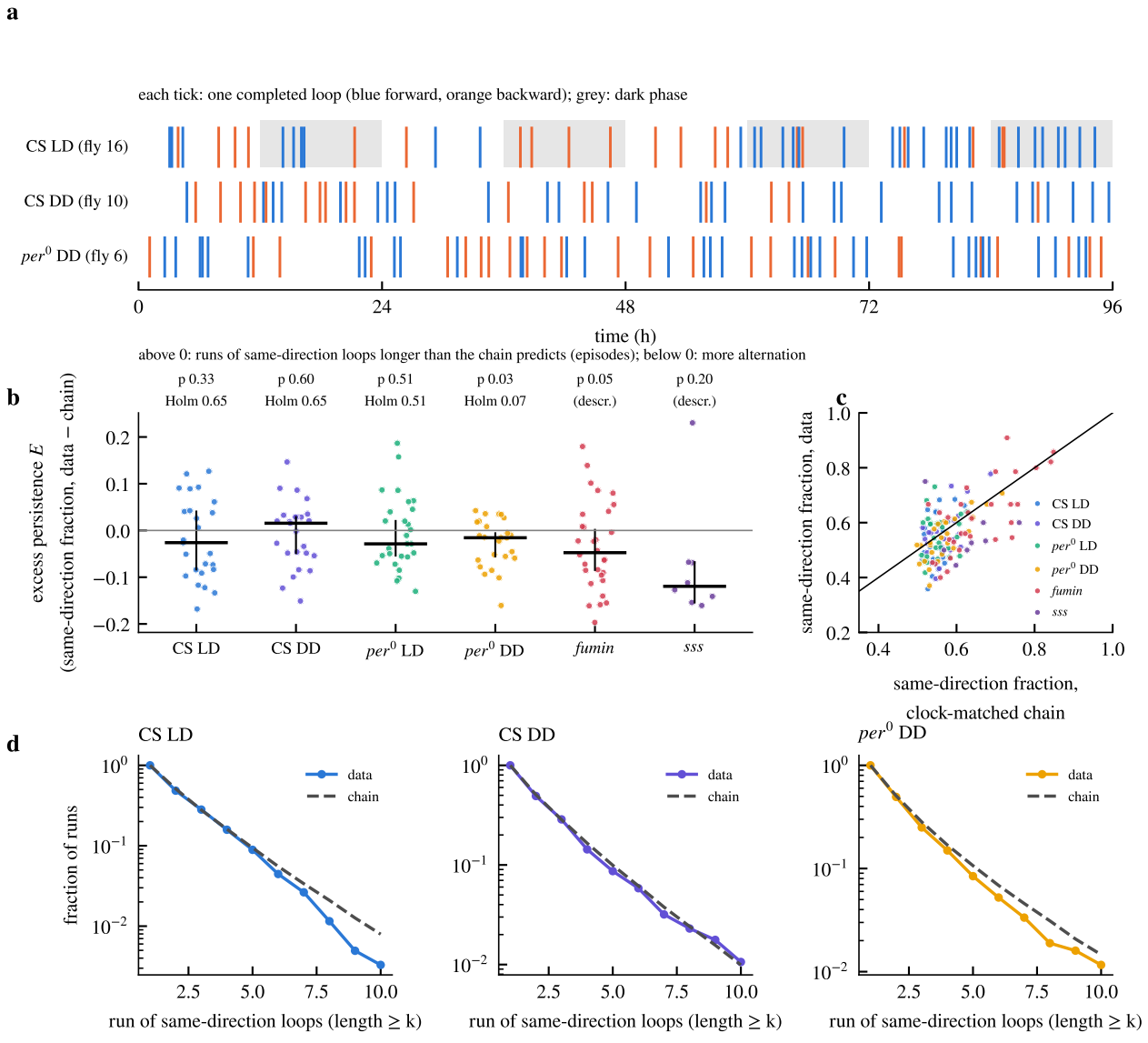
a
each tick: one completed loop (blue forward, orange backward); grey: dark phase
CS LD (fly 16)
CS DD (fly 10)
per0 DD (fly 6)
0
24
48
72
96
time (h)
above 0: runs of same-direction loops longer than the chain predicts (episodes); below 0: more alternation
b
p 0.33
Holm 0.65
p 0.60
Holm 0.65
p 0.51
Holm 0.51
p 0.03
Holm 0.07
p 0.05
(descr.)
p 0.20
(descr.)
excess persistence E
(same-direction fraction, data − chain)
0.2
0.1
0.0
−0.1
−0.2
CS LD
CS DD
per0 LD
per0 DD
fumin
sss
c
same-direction fraction, data
1.0
0.8
0.6
0.4
0.2
0.4
0.6
0.8
1.0
same-direction fraction,
clock-matched chain
CS LD
CS DD
per0 LD
per0 DD
fumin
sss
d
CS LD
CS DD
per0 DD
data
chain
fraction of runs
10^0
10^−1
10^−2
2.5
5.0
7.5
10.0
run of same-direction loops (length ≥ k)